

Transformation Protocol (M0370)

Overview

Chemically competent strains of *E. coli* (commercially available or prepared by user) can be transformed by ligation products prepared using the Instant Sticky-end Ligase Master Mix. Electrocompetent cells are not compatible. Users of competent cells from other vendors may need to dilute ligation reactions 4-fold, prior to transformation, in order to achieve maximum transformation efficiency. Not all cells from other vendors will benefit from this additional step. The following protocol is recommended by NEB. Other protocols can be used but the volume of ligation reaction used should not exceed 5 μ l reaction per 50 μ l cells.

Protocol

1. Thaw competent cells on ice.
2. Aliquot 50 μ l of cells into a 1.5 ml microcentrifuge tube.
3. Add 2 μ l of the ligation reaction to the cells and mix by finger-flicking. Do not vortex the tube.
4. Incubate the tube on ice for 30 minutes. Do not mix.
5. Heat shock at 42°C for 30 seconds, return tube to ice for 2 minutes.
6. Add 950 μ l recovery media (e.g. SOC) to the tube and incubate for one hour at 37°C with rotation or shaking (200–250 rpm).
7. Spread 100 μ l of the outgrowth (undiluted or diluted 1:5 with recovery media) onto appropriate antibiotic selection plates and incubate overnight at 37°C.

8. **Typical Results:**

Transformation efficiencies around 2×10^6 cfu/ μ g are typically achieved for recombinant cohesive end substrates (vector + insert), using cells with a 7×10^8 calculated efficiency with uncut DNA. This corresponds to several hundred colonies on a plate when 100 μ l of a 1 ml outgrowth is plated at a 1:5 dilution. As with all ligation and transformation protocols, many factors affect the calculated transformation efficiency, including purity and integrity of DNA ends, competence of the cells being transformed, media choices, incubation temperatures and times and biological effects (intact ORF in high-copy vector, toxic genes, etc.).